Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years shipping **TypeScript**, **Node.js**, **React**, and **MongoDB** products on **Google Cloud** and Azure, with authentication, RBAC, and multi-tenant security experience and an IEEE identity paper.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a **multi-tenant** platform on Azure SQL and Blob Storage with **row-level security** isolating tenant data for 500+ users, cutting dashboard latency from 1.2s to 350ms through indexed query optimization.
- Restructured fault-tolerant **async LLM orchestration** with background workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs and tripling throughput under concurrent load without cascading task failures.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a full-stack **React** and Flask grading platform with Flask-Login **authentication** and a switchable OpenAI, Anthropic, and Qwen model layer, serving 150+ instructors and 10K+ daily requests.
- Designed secure PDF, DOCX, and ZIP ingestion with concurrent batch processing through **REST APIs** and AWS RDS, handling 1K+ files per batch while cutting feedback turnaround 60% for instructors.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Engineered cross-platform data-sharing software on a **microservice architecture** with Jenkins CI/CD and 20+ unit and integration test suites, collaborating in Agile sprints with Microsoft and GitHub stakeholders.
- Optimized large-scale **API integrations** with object-oriented Singleton and Factory patterns, pagination, field filtering, and caching, improving response times 30% for low-bandwidth users and reducing technical debt 20%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **MERN** platform in **TypeScript** with eight Node.js microservices for authentication, resources, billing, leads, and notifications, syncing 2M+ MongoDB records across locations via REST APIs and WebSockets.
- Implemented **JWT** authentication and **role-based access control** services, deployed Docker builds to Google Cloud through Jenkins CI/CD with Snyk dependency scanning, reaching 95%+ build validation across releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Implemented **wallet-signature authentication** and JWT sessions for passwordless login, eliminating password storage for wallet users while securing booking and payment APIs built with Node.js, Express, and TypeScript.
- Designed the **end-to-end architecture**, mapped 12+ integration points, and presented it to the client, then delivered booking APIs with Firebase real-time sync behind React and React Native clients for 50K+ bookings.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, HTML, CSS
Full Stack: React, Next.js, React Native, TailwindCSS, Node.js, Express.js, REST APIs, WebSockets, Microservices, FastAPI, Flask
Databases: MongoDB, PostgreSQL, Azure SQL, MySQL, Firebase
Cloud & DevOps: Google Cloud Platform, AWS, Microsoft Azure, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD
Identity & Security: Authentication, JWT, OAuth 2.0, RBAC, Row-Level Security, Multi-Tenant Isolation, Clerk, Snyk
Practices: System Design, Design and Code Reviews, Agile/Scrum, Stakeholder Communication

Projects

Machine Identity Lifecycle Manager | *TypeScript, Node.js, React, MongoDB, Google Cloud, OAuth 2.0*

- Designed a **TypeScript/Node.js** service inventorying service accounts, API keys, and AI-agent credentials from GCP, with MongoDB-backed ownership records, scheduled rotation, **least-privilege RBAC** policies, audit logs, and a React console.

Comprehensive Analysis of Blockchain for User Authentication Paper | *RBAC, Smart Contracts, Self-Sovereign Identity*

- Co-authored an IEEE paper evaluating blockchain authentication for online users, **IoT device identity**, and role-based access control, including self-sovereign identity, smart-contract verification, and replay and phishing threat mitigation.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets, Kubernetes, TailwindCSS*

- Built a **TypeScript** and Next.js fintech platform on MongoDB with Plaid bank linking, a token-based firewall guarding transaction access, and WebSocket real-time updates across 1M+ transactions for 50K+ users.